Short Range Offline Communication System

A Dissertation (**MAJOR PROJECT PHASE – I)** submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS & COMMUNICATIONS ENGINEERING

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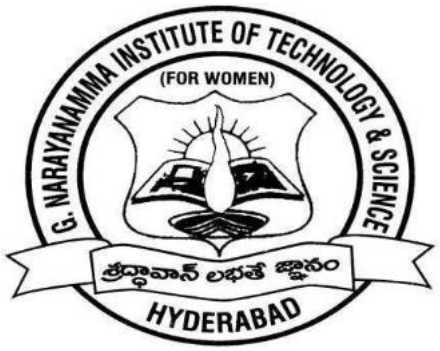
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INDEX

S. No.	Topic	Page No.
	Abstract	i
	List of Figures	ii
	List of Tables	iii
	List of Acronyms	iv
1.	Introduction	1
2.	Literature Survey	5
3.	Problem Statement and Objectives	9
	3.1 Problem definition	9
	3.2 Objectives	11
	3.3 Conclusion	13
4.	Proposed System	15
	4.1 Block Diagram	15
	4.1.1 Block Diagram (Zigbee + Bluetooth)	15
	4.1.2 Block Diagram (Zigbee + Wi-Fi)	18
	4.2 Proposed System	19
	4.3 Workflow	22
5.	Hardware Components	25
	5.1 Arduino Uno	25
	5.2 ESP8266	26
	5.3 LCD Display	27
	5.4 HC-12 Zigbee Module	28
	5.5 Li-ion Battery	29
	5.6 Voltage Regulator	30
	5.7 Supporting Components	31
	5.8 Conclusion	32
6.	Implementation	33
	6.1 Results	34
	6.2 Conclusion	37

6.3 Future Scope 38

7. References 40

ABSTRACT

During disasters, traditional mobile networks may fail, making communication impossible for victims. To overcome this, an emergency public safety communication system is proposed to send digital text messages from mobile phones to a wireless communication system that transmits the information to a remote monitoring station.

This two way communication system, built using Arduino, LCD, Wi-Fi, Bluetooth, and Zigbee modules, can be installed in disaster-prone or high-risk areas like coastal villages, hospitals, colleges, and industries.

It helps transmit messages even in network black spots (areas without mobile coverage) using Zigbee for long-range communication. When a message is received, a beep alert notifies the user. The prototype demonstrates reliable text communication between distant systems without depending on mobile networks.

LIST OF FIGURES

S. No.	Figure No.	Figure Name	Page No.
1	4.1	Block Diagram (Bluetooth + Zigbee)	15
2	4.2	Block Diagram (Wi-Fi + Zigbee)	18
2	5.1	Arduino Uno	25
3	5.2	ESP8266	26
4	5.3	LCD Display	27
5	5.4	Li-ion Battery	29
6	6.1	Complete Circuit	33
7	6.2	Victims Message	35
8	6.3	Rescuer’s Response	36

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LIST OF TABLES

S. No.	Table No.	Table Name	Page No.
1	5.1	ESP8266	27
2	5.2	LCD Display	28
3	5.3	Zigbee Module	2
4	5.4	Li-ion Battery	30
5	5.5	Voltage Regulator	31

ACRONYMS

S. No.	Abbreviation	Full Form / Description
1	GND	Ground
2	IOT	Internet of Things
3	LCD	Liquid Crystal Display
4	MCU	Microcontroller Unit
5	RF	Radio Frequency
6	SOS	Save Our Souls
7	Wi-Fi	Wireless Fidelity

CHAPTER 1

INTRODUCTION

Communication is the most basic human need in society. It is also the most important need in emergency situations and natural disasters. It helps to facilitate the coordination of people, rescue teams, and authorities to save lives and utilize resources optimally. However, in natural disasters such as cyclones, floods, earthquakes, or tsunamis, communication through mobile towers, internet services, and wired telephone lines is impossible due to damaged infrastructure, power cuts, or excessive communication channel congestion. Therefore, the victims of natural disasters cannot communicate their location to the emergency response teams. This leads to a delay in emergency response operation, causing more damage to the natural disasters.

To resolve this problem of communication failure in emergency situations, a short range offline communication system (OECS) is proposed. This will provide a reliable offline communication facility to the victims of natural disasters through wire-free communication. This will enable the victims to send digital text messages to emergency response centers without relying on the internet or mobile communication services. The proposed system will be designed using an Arduino Uno microcontroller with Bluetooth, Wi-Fi, and Zigbee technologies to facilitate seamless wire-free communication.

In traditional mobile communication systems, black spots or dead spots are defined as those areas where mobile communication signals are not available or are very low. Black spots are usually found in remote village areas, mountainous regions, underground tunnels, or coastal areas where mobile towers are not able to provide connectivity. In the event of a disaster, these spots are more important since communication links are completely lost. To overcome this drawback, the proposed system uses Zigbee transceivers that are able to provide two-way wireless communication without any external network support. In this way, an independent network is provided to allow effective communication in disaster recovery efforts.

The proposed Offline Emergency Communication System includes four communication points with an Arduino Uno controller, LCD display device, Zigbee module, Wi-Fi module, Bluetooth module, buzzer device, and power supply. The first communication point receives a text message from a user's mobile phone via Wi-Fi or Bluetooth technology. The message will

then be processed by the Arduino controller, which will send the message to the next communication point through a Zigbee transceiver module. At the next communication point, the Zigbee will receive the message, which will display the message on an LCD screen while at the same time sending a beep sound as a notification to the users that they have a new message. This will, therefore, send a message from the next communication point, which will be sent back through the same process, thus creating a two-way communication channel.

The Short Range offline Communication System (OECS) is designed to address this problem through the provision of uninterrupted two-way communication during disasters. The system allows users to send digital text messages from a mobile phone to another nearby wireless communication system. The message is then sent from the communication system to another remote communication point. This helps to facilitate communication between victims of disasters and rescue teams even in the absence of communication infrastructure. The system is completely offline using Wi-Fi technology and Zigbee modules in combination with an Arduino Uno microcontroller for the efficient transmission of messages.

In many areas of the country, especially in rural areas, there are areas that are considered black spots where there is either poor or no mobile communication network. Black spots are usually experienced in coastal areas, hill areas, underground tunnels, industrial areas, etc. During disasters, these areas act as a major impediment to communication. The proposed system is helpful for these areas as Zigbee technology is utilized for communication. Zigbee technology is capable of creating a wireless network that covers reasonable distances without the need for a mobile tower. Zigbee technology is considered secure for transferring information between different nodes.

The system architecture of the proposed system will comprise four points of communication, each of which will have an Arduino Uno, LCD display, Wi-Fi/Bluetooth, and Zigbee. The user will be able to send a message from a mobile phone to the first point of communication, i.e., Point A, using either Wi-Fi or Bluetooth. The Arduino will be able to read this message and send it to another point of communication that is farther away using the Zigbee protocol. The user will be alerted by a beep sound from the buzzer connected to the LCD display. This process will be repeated to send a message from another mobile phone connected to the second point of communication, thus enabling two-way communication. This method will be

highly effective for the purpose of sending short emergency messages when mobile communication is not available.

The hardware components of the system are the microcontroller of the Arduino Uno, which acts as the CPU of the system. The ESP8266 Wi-Fi module will be used for wireless message transfer. The Bluetooth module will be used to establish a connection with the mobile phone. The Zigbee transceivers will be used for point-to-point communication. The LCD display will be used to display the messages, while the buzzer will be used to send alerts to the user. The power supply unit will be used to provide power to all the components of the system. This will be a lightweight system that will be highly portable.

This offline system does not require any mobile towers or internet connection, which makes it more suitable for use in disaster zones or even in remote areas. The beep alert is also used as a form of immediate alert, which will definitely grab the user's attention in case of emergencies. The Zigbee protocol has been used in the prototype, but it can be extended further by adding more nodes. The proposed system can be used in different field applications due to its small size, portability, and low cost. The system can be implemented in college campuses, hospitals, industries, shopping malls, and even in coastal villages, as safety is a primary requirement in these areas. The use of Wi-Fi, Bluetooth, and Zigbee has been integrated in the system, which provides flexibility. The system can be used even in case of failure of any module.

The Arduino Uno microcontroller has been used as a reliable interface between the user and the communication module. The message can be processed and transmitted using the Arduino microcontroller. The software tools used in the project are MIT App Inventor, Arduino IDE, and XCTU Software. The MIT App Inventor has been used in the project, which is used to create a simple Android application using which the user can send and receive text messages using Bluetooth or Wi-Fi. The Arduino IDE has been used in the project, which is used to program the microcontroller. The XCTU software has been used in the project, which is used to configure the Zigbee transceivers.

The use of Arduino Uno as the primary processor allows for ease and stability. Arduino is open-source, which allows it to be used in different communication protocols. This makes it easier and more efficient in integrating different wireless communication modules. The device processes the information received, processes it accordingly, and controls all output processes. This includes the display of messages on the LCD screen and the activation of the buzzer.The

modularity also allows room for improvement, which can be used in enhancing the real-time response capability. The efficiency in power usage and portability allows the system to operate using rechargeable batteries in the absence of a power supply.

The Offline Emergency Communication System is not only limited to the usage of communication in the event of a power outage; it also has the potential to serve as a public safety system. It can also be installed in critical zones such as hospitals, industrial plants, educational institutions, and coastal zones. It can also be used to send messages in the event of emergencies. It can also be installed in underground mines, remote areas, and military field operations. This is because it does not rely on cellular and internet connectivity.

On the whole, the proposed project of the Short Range Offline Communication System is a unique solution that can be considered viable for ensuring communication in emergency situations even when other communication channels are down. With the help of Arduino Uno, Zigbee, Wi-Fi, and Bluetooth technologies, the proposed system ensures the continuation of communication between the victims and the rescue parties without relying on other communication channels. Not only is the efficiency of the rescue operation guaranteed by the proposed system, it is also more beneficial in ensuring the safety of the general public because there is no chance of keeping anyone disconnected in emergency situations. Thus, the proposed project is a reliable communication system that can play a vital role in saving more lives in emergency situations.

CHAPTER 2
LITERATURE SURVEY

[1] The objective of this research is to create a wireless communication system using Wi-Fi technology to send digital messages between users during an emergency. In this paper, the authors have proposed a system using ESP8266 Wi-Fi technology and a microcontroller to send short-range messages. In this system, wireless communication technology, i.e., Wi-Fi technology, is used to send wireless messages in real-time during an environment where mobile communication services are not available. In this system, it is possible to send a message from a smartphone to a wireless control system and display it on an LCD screen. The objective of this system is to create an independent channel to send wireless messages during an emergency.
Limitations: This system does not support long-range wireless communication due to the characteristics of the wireless communication technology used. This system will not support wireless communication if there are no Wi-Fi devices available in the environment. This system does not support wireless communication over a long range. This system does not support wireless communication between various nodes. This system does not support wireless communication using other wireless technologies like Bluetooth or Zigbee.

[2] In this paper, the authors are focused on developing a wireless communication system using Zigbee technology for wireless communication in disaster-prone areas. In this system, a network is created between multiple nodes using Zigbee technology for wireless communication. In this system, text messages as well as environmental information can be sent wirelessly between nodes. In this paper, the authors are focused on using Zigbee technology for wireless communication in rural areas as well as coastal areas in case of flood or storm conditions.
Limitations: In this system, Zigbee technology is used for wireless communication between nodes in a short range. However, Zigbee technology cannot be used for large-scale environments. In this system, there is no integration of mobile phones for sending text messages using those phones. There is also no alert provided to users to alert them when they receive text messages.

[3] This study has proposed a communication model through the usage of Bluetooth modules with Arduino microcontrollers for short-range message transmission. The system allows the user to communicate by sending messages from a mobile phone to another device with a Bluetooth module. The authors have also mentioned the advantages of using Bluetooth, which include low cost, low power consumption, and usage of mobile phones. The proposed model has depicted the efficiency of the device pair, reliable information transmission, and the usage of Android-based applications.

Limitations: The study has only considered the usage of Bluetooth technology for device connectivity. The study has not considered the usage of multiple nodes for communication. The system will not be useful during disaster situations, as communication will have to take place at a greater distance. The study has not considered the usage of other wireless communication modules like Zigbee or WiFi.

[4] This research has proposed a Wireless Sensor Network (WSN) framework with Zigbee transceivers and microcontrollers for monitoring and transmitting information during disaster-prone areas. The authors have proposed a system with Zigbee modules, which are connected as a network for transmitting vital information during disaster situations like temperature, humidity, and smoke levels. The major aim of the system is to send alerts regarding disaster situations.

Limitations: The study has mainly focused on sensor nodes, but it has not considered the usage of the system for sending information between users. The study has not proposed a mobile-based system through which a user can send text messages. The system has not provided a two-way communication system between the users and the monitoring station. The system has not provided a fixed position system, nor has it provided an audible system.

[5] This paper aims to present a hybrid model in which Zigbee and Wi-Fi modules can be used in offline public safety communication. The system allows users to send messages in text format by using their smartphone through the Wi-Fi/Bluetooth module. The messages are then sent to other remote stations using the Zigbee module. The design of the communication module is done using Arduino Uno, LCD, and buzzer technologies. The paper also emphasizes the importance of the system in remote areas such as the coast, hospitals, and colleges.

Limitations: The paper presents efficient communication technology in the form of an offline communication system. However, the paper lacks information on the efficiency of the system in terms of scaling up the system. The system is only efficient in short and medium-range communication using the Zigbee module. The paper also lacks information on the performance of the system in harsh environmental conditions.

[6] This paper presents a wireless emergency communication system based on the Internet of Things. The system uses microcontrollers and wireless transceivers such as Wi-Fi and Zigbee to send messages. This system is efficient in helping users share information in real time in disaster situations. The paper also presents how the integration of IoT devices can help users in data transmission in disaster situations. The paper also presents how messages can be sent between Arduino devices using wireless transceivers. This allows users to communicate with one another without the help of the internet.

Limitations: The system presented in the paper is efficient in short-range communication. However, the system lacks integration with mobile phones to help users access the system. The system mainly focuses on data transmission between two nodes. However, the system lacks integration with portable devices such as mobile phones. The system also lacks the feature of buzzers to alert users when they receive messages.

[7] The paper proposes a microcontroller-based offline communication network, which is suitable for rescue operations in disaster-stricken areas. The authors have employed Arduino Uno, Bluetooth, and Zigbee to establish short- and medium-range communication networks. The authors have sent messages from a smartphone to the microcontroller via Bluetooth, which is then sent to the receiving end via Zigbee. The authors have displayed the messages on the LCD screen and activated a buzzer.

Limitations: The authors have mainly concentrated on hardware implementation and short-range testing in a laboratory environment. The authors have not tested the effectiveness of the proposed system in real-time scenarios. The authors have not discussed the effectiveness of the proposed system in real-time scenarios. The authors have not discussed the effectiveness of the proposed system in terms of interference, power consumption, and environmental conditions, which are important in real-time scenarios. The authors have not discussed the effectiveness of

the proposed system in terms of interference, power consumption, and environmental conditions, which are important in real-time scenarios. The authors have employed Zigbee, which has limited range, making it unsuitable for disaster scenarios where disaster areas may be spread over a large area.

CHAPTER 3

PROBLEM STATEMENT

3.1 PROBLEM DEFINITION

For instance, in the case of natural disasters like earthquakes, floods, cyclones, tsunamis, building collapses, and industrial accidents, which are likely to cause severe destruction and loss of infrastructure and communication systems, communication among affected individuals and with rescuers is perhaps the most important factor in deciding survival. However, in most cases, due to the collapse of traditional communication systems like mobile towers, optical lines, and internet gateways, a complete communication blackout occurs. In such cases, individuals stuck in debris or in remote locations are not able to send any message to rescuers or convey their location to rescuers.

In the present world, most communication systems are based on internet or mobile communication systems. Such systems are heavily dependent on physical infrastructure and a constant source of power. In most cases of natural or human-induced disasters, these systems are likely to collapse due to a lack of power or network congestion. In such cases, most existing communication channels like GSM and Wi-Fi are likely to be unavailable. Although satellite phones can be used in such cases, they are not available to everyone in the world and are also very expensive.

The other significant limitation in the current disaster management system is that there is a lack of development of an independent, self-sustaining, and long-range wireless communication system that is capable of operating independently without the need for existing infrastructure. Manual rescue operations are usually carried out through visual or auditory cues to locate the victims of disasters. However, these methods are inefficient and prone to human error. Modern technologies such as thermal imaging, drone technology, and sound sensors assist in the rescue operations. However, these technologies face limitations in harsh environments, especially when the visibility is poor, the victims are unconscious, and there are obstacles to the transmission of sound.

The lack of development of an independent and low-power communication system also makes it impossible to maintain coordination between multiple rescue units operating at

different locations. In the case of large-scale disasters, there are usually multiple rescue units operating at different locations. In such cases, the lack of communication between these units leads to inefficient utilization of resources.

Furthermore, rural and coastal areas are also more prone to other risks. In rural and coastal areas, mobile network connectivity is already low in normal conditions. In disaster conditions, these places are completely cut off from mobile network connectivity. For instance, fishermen working deep in the sea or people staying in islands cannot depend on mobile network connectivity for their safety. Similarly, in underground places like mines, bunkers, or tunnels, radio frequency signals cannot penetrate. Therefore, people working in these places are cut off from the external world in disaster conditions. These examples show that what is required is a strong wireless communication system that can function in all terrains.

In order to overcome these challenges, which are still existing in wireless communication systems in disaster conditions, a Short Range offline Communication System is proposed. This system is low in cost, portable, and energy-efficient and can function in the absence of mobile network connectivity. The proposed system uses Arduino Uno as its central processing unit and combines it with Zigbee (HC-12) and Wi-Fi (ESP8266) technology to form an ad hoc network. This system facilitates both ways of communication between survivor units and rescue units.

The survivor unit, which is operated by a rechargeable Li-ion battery, allows individuals to transmit pre-defined messages to rescue teams using push buttons or a Bluetooth-enabled smartphone application. The messages are wirelessly transmitted to the rescue team using the Zigbee or Wi-Fi module. The rescue unit has another Arduino controller that receives messages, processes them, and displays them on an LCD screen. It sends a buzzer alarm to the rescue team at the same time.

Unlike the traditional communication system that uses a lot of power, this offline communication network is independent and easy to deploy. This communication network allows for real-time communication, low power consumption, and reliability, which are three key factors in any communication network.

The module has been designed in such a way that it can be easily expanded to include GPS, LoRa, and health sensors, which will make it even more powerful.

In other words, the proposed Offline Emergency Communication System is a direct solution to the limitations of the existing communication infrastructures since it offers a self-sufficient, reliable, and deployable communication system that can be utilized during emergency situations. It enables both the victims of disasters and rescuers to communicate effectively even in situations where there is a total network outage.

3.2 OBJECTIVES

1.Wireless Communication During Emergencies

The major aim of this project is to design a wireless communication system that will be effective even during emergency situations when other communication channels, like mobile communication networks and the internet, are not available. During natural calamities like earthquakes, floods, and cyclones, communication channels like mobile towers and power lines are usually damaged, which causes the isolation of the victims of the disaster. In order to resolve this issue, the proposed system will be designed in a manner that will create a self-contained network, which will allow communication even when other communication channels are not available. The system will use Arduino Uno as a controller, along with other wireless communication systems like Zigbee (HC-12) and Wi-Fi (ESP8266), which will allow communication between the survivor units and the rescue units. The survivor units will send an SOS message using buttons or Bluetooth technology, while the rescue units will receive the SOS messages using an LCD screen with a sound from a buzzer. Thus, through this project, the major aim of ensuring communication even during life-threatening emergencies will be met.

2.Support for Rural and Public Environments

One of the major objectives of this project is to design a system that can operate in rural, public, and coastal environments. This is because these environments usually have poor network coverage. Many rural environments usually lack stable communication facilities. This makes such environments very prone to disasters. The system to be designed in this project is one that offers a wireless network. This wireless network is one that can operate independently without the availability of the internet and mobile networks. This way, communication in a given area can always be guaranteed.The system to be designed in this project utilizes an Arduino Uno microcontroller. This microcontroller processes data and transmits it to other devices using Zigbee and Wi-Fi modules. This way, a wireless

communication system is created. This system is very useful in environments such as villages, schools, hospitals, factories, and in environments where there is usually a large number of people. This way, a large number of people can be saved in case of emergencies. This objective ensures that both rural and public environments can operate independently. This way, communication in such environments can always be guaranteed.

3.Rescue Team Assistance

The third objective is to enable fishermen and other workers at sea to have access to reliable communication services in areas where there is no access to a mobile network. On the open waters of the sea, there is no access to cellular communication beyond the shores. This puts fishermen at great risk of losing their lives in case of failure of their machines, during a storm, or in case of accidents. The proposed system will provide the fishermen with access to offline communication services through Zigbee technology (HC-12), enabling the transmission of data up to 1 km. The Arduino Uno will be used to control the system to send SOS messages from the fisherman's device to the coastal station. The system will be powered using a Li-ion battery to enable operation even without power connection. This objective will provide fishermen with access to life-saving communication services that will promote their safety at sea.

4.Communication for Islands and Isolated Areas

This objective is specifically designed to enable effective communication in island regions or isolated areas where there is a lack of telecommunication facilities. These regions experience total disconnection in case of a calamity like a tsunami or a cyclone. The proposed Offline Emergency Communication System includes a Zigbee (HC-12) and Wi-Fi (ESP8266) module, which enables the devices to communicate among themselves in the coverage area of the island. The Arduino Uno board processes the information and sends it in real-time among multiple devices. The LCD and buzzer provide a visual and sound alert. The system operates on rechargeable batteries, which can be used for a long period without electricity. The residents can send distress signals using the button input or a Bluetooth mobile application. This objective is designed to provide self-reliance in communication even in isolated island regions.

5.Connectivity in Underground Structures

The final aim of the system is to ensure communication in areas like mines, tunnels, and bunkers, which have poor mobile and radio communication signals. The people working in these areas

are usually at greater risk, and if there is an emergency, the lack of communication facilities might result in severe damage. The proposed system will help overcome this problem with the use of Zigbee (HC-12), which will provide better communication, as its signals have better penetrative abilities compared to normal high-frequency signals.

Every worker will have a transmitting device through which emergency signals will be sent to a receiving device located above the ground. The receiving device will have an Arduino Uno, an LCD display, and a buzzer, which will immediately send signals to the supervisor regarding an emergency. This aim will ensure that the people working underground are never isolated, as emergency signals will reach the above-ground control units immediately.
It will help create a safer working environment with the help of technology, even in extreme working conditions.

3.3 CONCLUSION

In conclusion, the problem statement clearly indicates that there exists an urgent need for establishing a reliable means of communication that will always be available during calamities or emergencies when other means of communication are out of order. Situations like cyclones, earthquakes, floods, and infrastructures out of order result in a complete blackout of communication. This means that the victims will not be able to send out messages, while the rescue teams will not be able to get vital information. The Short Range Offline Communication System will completely eliminate this problem, as it will provide an independent, self-sustaining, and easy means of communication without depending on mobile networks, the internet, or other means of communication.

The microcontrollers, wireless, and other electronic components will provide a smooth short-range data transmission between victims and rescue teams. The use of portable battery power will provide a continuous power supply even during calamities or emergencies when other means of communication are out of order. The use of Li-ion batteries with a protection circuit will not only provide power efficiency but will also be safe and long-lasting, which is required during rescue missions.

The ability to send SOS messages, location, as well as status, could be helpful to rescue teams. This would allow them to have better awareness of the situation, which could lead to quicker decision-making. The durability of this system to work under harsh conditions makes this a valuable asset to disaster recovery.

In conclusion, the Short Range Offline Communication System is an impressive technology solution as it is practical, innovative, and life-saving. The role that this system plays in ensuring that there is uninterrupted communication between victims and rescue personnel is invaluable. This would greatly enhance disaster preparedness. This project not only fills a critical gap in technology but also plays an important role in building a stronger community that is able to cope with disasters.

CHAPTER 4

PROPOSED SYSTEM

4.1 BLOCK DIAGRAM

4.1.1 BLOCK DIAGRAM (BLUETOOTH + ZIGBEE UNIT)

The above block diagram represents the design of the Offline Communication System using the major hardware components of the Bluetooth and Zigbee technology. The Arduino Uno microcontroller is the main processing unit of the system. It is connected to other hardware components such as the Bluetooth module for user input, Zigbee (HC-12) for long-distance communication, the display unit, and the power supply unit to complete the offline wireless communication system. The system is designed to work properly even without the availability of mobile networks and internet access.

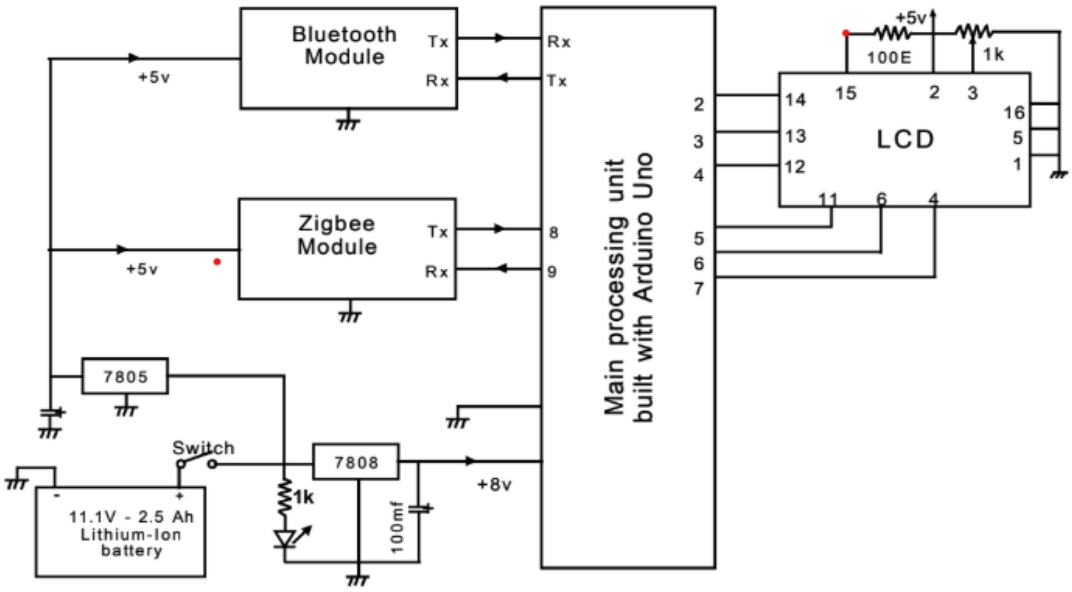


Fig 4.1 Block Diagram(Zigbee + Bluetooth)

This section mainly deals with the integration of the Bluetooth and Zigbee protocols for short-range and long-range communication, respectively. The main components of this block diagram are the Arduino Uno, Bluetooth module, Zigbee module, LCD display, voltage regulator, and 11.1V Li-ion battery, which acts as the power source for this circuit.

Main Processing Unit

The main component of this circuit is the microcontroller unit, which controls all the operations of the circuit. The microcontroller unit of this circuit is the Arduino Uno. The Arduino Uno microcontroller unit processes the data received from the Bluetooth module to send and receive messages using the Zigbee protocol. The Arduino Uno uses its serial pins to communicate with the Bluetooth module to send and receive messages. The Arduino Uno uses its digital pins to communicate with the Zigbee module to send and receive messages. The Arduino Uno also controls the LCD display to show the output of the circuit. The Arduino Uno controls the buzzer to show the alerts.

Communication Modules

Two types of wireless modules are used in this system to establish proper communication. These are as follows:

1. Bluetooth Module (HC-01)

Bluetooth technology is used in this project to interface the user with the system. This module allows a smartphone to directly connect with the Arduino microcontroller using wireless communication. The user will be able to send messages to other people using a mobile application after pairing with the HC-05 module. This module will work on 5V voltage and will communicate with the Arduino microcontroller using serial communication on Tx and Rx pins. This module will allow users to send messages without using internet connectivity.

2. Zigbee (HC-12) Module

The Zigbee module is used to facilitate wireless communication between different nodes of the system. It transmits and receives encoded data packets wirelessly within a range of 1 km. The Zigbee module is connected to the Arduino using digital pins and is given a 5V supply. After receiving the message from the Bluetooth module by the Arduino board, the message is sent to the Zigbee module to transmit it to the receiver unit.

Display Section

The 16x2 display is used to connect to the Arduino board to display the output of the incoming and outgoing messages in real time. It is used to display information to the users. The display is given a 5V supply by passing it through a resistor to control the flow of current to the display to adjust the contrast. This helps the rescue team to read the messages easily.

Power Supply Section

The power source for the system is provided by a 11.1V 2.5Ah Lithium Ion battery. Voltage regulators are utilized to maintain different voltage levels for the components. The 7808 voltage regulator is utilized to step down the battery voltage to +8V. Then the 7805 voltage regulator is utilized to step down the +8V to +5V. This +5V is utilized to power the Arduino, Bluetooth module, Zigbee module, and LCD. Capacitors are utilized to maintain the voltage at a stable level. A switch is utilized to control the power, and an LED is utilized to indicate the status of the system.

System Function

When the system is turned ON, the Arduino initializes the components. The user sends an emergency message from their smartphone using the Bluetooth module. The Arduino gets the message and forwards it to the Zigbee module for long-distance transmission. The receiver unit gets the signal from the Zigbee module, decodes the message, displays it on the LCD, and turns on a buzzer to alert the rescue team.

Conclusion

The above block diagram shows an efficient offline communication system that uses Bluetooth for user interaction and Zigbee for long-range data transmission. This system helps to facilitate efficient and uninterrupted communication during critical times when network connectivity is not available.

4.1.2 BLOCK DIAGRAM (Wi-Fi + ZIGBEE UNIT)

The block diagram shows an overview of the hardware components that are used to develop the Offline Communication System using Wi-Fi and Zigbee. This system uses an Arduino Uno microcontroller as the main processor unit. This system also uses an ESP8266 Wi-Fi module for wireless communication, a Zigbee module for long-range data transmission, a display unit, and a regulated power supply unit. This system uses an Arduino Uno microcontroller that integrates a wireless network interface to facilitate efficient communication during offline times.

The Offline Communication System uses Wi-Fi and Zigbee technologies to facilitate user interaction and long-range data transmission. The hardware components that are integrated into this part of the system are represented in the block diagram.

Main Processing Unit

At the heart of this system is the microcontroller. The microcontroller used in this system is the Arduino Uno. The microcontroller is responsible for controlling all the processes in the system. The microcontroller receives data from the Wi-Fi module, processes it, and then sends it to the Zigbee module to transmit the data. The microcontroller communicates with the ESP8266 module through the serial Tx and Rx pins. It controls the output devices such as the LCD display and the buzzer. It also ensures that the messages sent to the communicating nodes are formatted and encoded correctly.

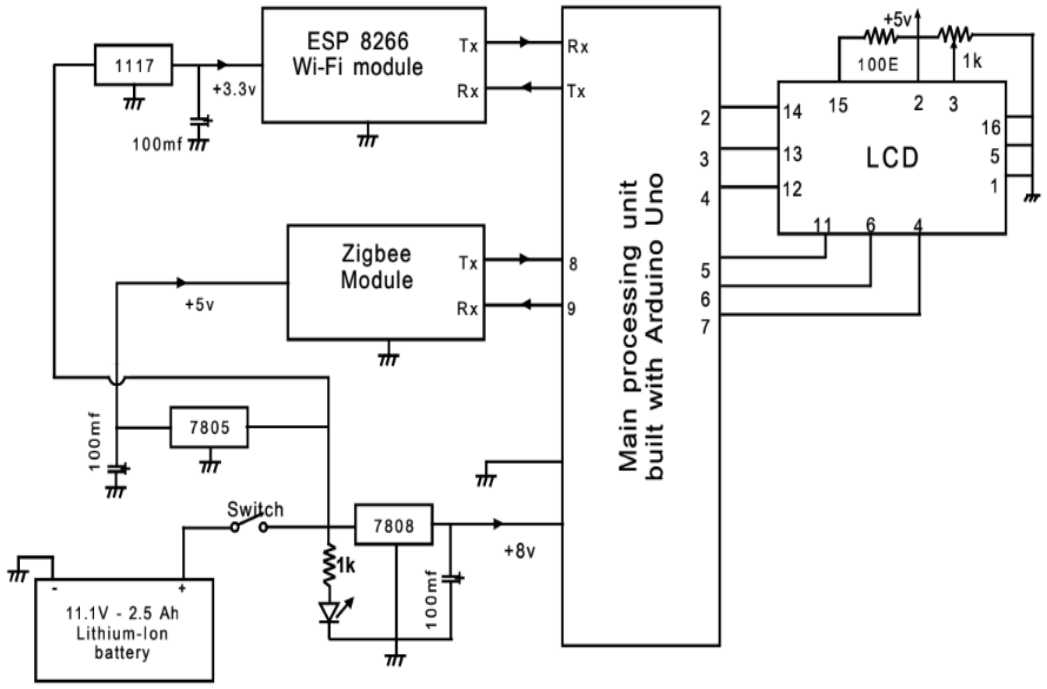


Fig 4.2 Block Diagram(Zigbee + Wi-Fi)

Communication Modules

Two wireless communication modules are used to facilitate communication as follows:

1. ESP8266 Wi-Fi Module

The ESP8266 module is used to establish a local network that enables smartphones to connect to the system without requiring internet connectivity. This module has the capability to work as an Access Point (AP), which enables users to connect to the system directly. This allows users to send messages using a web interface. This module works at 3.3V, which is provided by the LM1117 voltage regulator. This module uses serial communication with the Arduino board and is used as an alternative means to send emergency messages.

2. Zigbee (HC-12) Module

The Zigbee module facilitates long-range communication between different units of the system. The Zigbee module sends and receives messages wirelessly using radio frequency signals. The Zigbee module works at a supply voltage of 5V, which is provided through the digital pins of the Arduino. Once the message is processed by the Arduino, it sends the message to the Zigbee module for transmission.

Display Section

To display messages, a display unit of 16×2 LCD has been used. The display unit works at a regulated supply of 5V, which is provided through the Arduino. The display unit will help the rescue team read the messages sent through the system.

Power Supply Section

The system uses an 11.1V Lithium-ion Battery as its power source. Voltage regulators are used to maintain a constant voltage level in the system. The voltage regulator 7808 reduces the voltage level from 11.1V to 8V, which is further reduced by the voltage regulator 7805 to 5V, which is used by the Arduino, Zigbee, and LCD. The voltage regulator 1117 provides a constant voltage of 3.3V, which is used by the ESP module. The capacitors provide stability to the power supply. A switch and LED are also used.

System Function

When the Arduino is turned ON, it initializes all the modules. The user connects to the Wi-Fi hotspot provided by the ESP8266 module and sends the emergency message using the web interface provided. The Arduino receives the message and sends it to the receiver using the Zigbee module. The receiver system receives the message and displays it using the LCD and alerts the user using the buzzer.

Conclusion

The above block diagram represents a hybrid communication system that uses both Wi-Fi and Zigbee technologies. This system provides reliable communication for users in areas where there is no existing network infrastructure.

4.2 THE PROPOSED SYSTEM

The proposed system aims to establish a reliable means of communication that will be available offline during emergencies or disaster situations when communication networks such as mobile phones or internet connections are not available. This will allow people affected by disasters to

send messages to rescue teams using various wireless communication technologies. The proposed system has three components: the Mobile Interface, the Survivor Unit or Transmitter, and the Rescue Unit or Receiver. These components work together to allow seamless communication.

1. Mobile Interface

The mobile interface acts as the main point of interaction between the user and the communication system. The mobile interface provides users in distress the opportunity to send emergency communications quickly using their mobile device without the need for an internet connection. The system can communicate using both Bluetooth and Wi-Fi, which provides it with flexibility and reliability in different situations.

The mobile device, using the HC-05 Bluetooth module, can communicate directly with the system. Once connected, the user can send predefined SOS messages or any other message using a simple mobile application. This form of communication is efficient, quick, and can be used in situations when immediate communication is needed.

In addition, the system also uses the ESP8266 module, which provides Wi-Fi communication using Access Point mode. The Access Point mode provides a hotspot, which users can use to connect to the system. Once connected, users can use a web interface on their mobile device to communicate. The use of both Bluetooth and Wi-Fi provides the system with reliability in case one form of communication fails.

2. Survivor Unit (Transmission Unit)

The Survivor Unit is the transmitting component in the system. It is responsible for receiving, processing, and transmitting emergency messages over a long distance. The unit is based on the Arduino Uno microcontroller. This microcontroller is the central processing unit in the system.

The messages transmitted by the mobile interface using Bluetooth and Wi-Fi protocols are received by these components and transmitted to the Arduino microcontroller through serial communication. The Arduino microcontroller processes the messages by verifying and structuring the messages to ensure they are in the correct format. This way, the messages can easily be understood at the receiving end.

After the messages have been processed by the Arduino microcontroller, they are transmitted to the disaster area using the HC-12 Zigbee module. This module allows the messages to be

transmitted over a long distance. This is particularly important in disaster situations when communication lines are destroyed. The unit uses a rechargeable Li-ion battery to operate independently. This way, the unit can operate without relying on any other power source. The unit is portable to ensure that it can easily be carried in real-time situations.

3. Rescue Unit (Receiver Unit)

The Rescue Unit acts as a receiving end of this system. The main task of the Rescue Unit is to receive, interpret, and display the message sent by the Survivor Unit. The components used in this system are an Arduino Uno microcontroller, an HC-12 Zigbee module, a 16x2 LCD, and a buzzer.

The Zigbee module is always in receive mode. The main objective of being in receive mode is to receive signals from the Survivor Unit. Once a signal is received, it is forwarded to the Arduino controller. The Arduino controller then decodes the message and extracts the information sent along with the message. The information includes the identification of the users and the type of emergency. The information is then forwarded to the LCD, which provides a clear understanding of the situation. At the same time, a buzzer is also activated, which produces a sound and thus provides instant information of the message sent by the Survivor Unit.

System Overview

The proposed system will provide offline wireless communication, real-time alerts, as well as portability during emergency situations when other communication infrastructure is not readily available. The proposed system has the ability to work independently without the need for a GSM network or internet connectivity. This will be most beneficial during emergency situations when the location of the emergency is far from cities, disaster areas, as well as areas without internet connectivity.

The proposed system utilizes the potential of Arduino Uno as well as ESP8266 for efficient communication through the use of Wi-Fi as well as Bluetooth communication technologies. The proposed system will have better communication through the use of long-range wireless communication technology. This will be most beneficial during emergency situations when communication infrastructure is not readily available.

The proposed system will provide a number of benefits, including:

Reliable communication through access point mode as well as local connectivity mode

Low power consumption

Instant alerts through LCD display as well as buzzer

Easy usage with minimal technical knowledge

In conclusion, it can be said that the proposed system provides a self-sustaining and reliable system of emergency communication, which allows uninterrupted data communication between the affected individuals and rescue teams. The portability, scalability, and independence of the proposed system from other external networks make it a viable solution in today's disaster management and emergency response applications.

4.3 WORKFLOW

The Short range Offline Emergency Communication System operates on a workflow, which provides real-time communication between the affected individuals and rescue teams even in the absence of conventional communication systems such as GSM or the internet. The workflow involves three main stages.

1. System Initialization

The workflow of the Offline Emergency Communication System starts with the initialization of the Offline Emergency Communication System. This initialization occurs as soon as the power is turned on using the rechargeable Li-ion battery. At this point, the Arduino Uno microcontroller initializes all the hardware components that are connected to the system. This is to ensure that all the components are functioning properly. At the same time, the Bluetooth module (HC-05) is initialized to pairing mode to enable it to connect to a smartphone. The Wi-Fi module (ESP8266), if enabled, is initialized to access point mode to enable it to connect to a smartphone. The Zigbee module (HC-12) is initialized to the appropriate frequency to enable it to transmit and receive information. At the same time, the LCD is initialized to display system status messages such as "System Ready" or "Waiting for Input." Additionally, the buzzer may make a short sound to indicate that the system is functioning properly. This is the most important step in the workflow of the Offline Emergency Communication System because it ensures that the system is functioning properly without any delay.

2. Message Generation and Input

After the system has been initialized, the user can now send an emergency message using the process, which will be created and sent using a mobile device, which will act as an interface for the user. The mobile device will be connected to the system using a smartphone, which will use

Bluetooth technology or Wi-Fi, depending on their availability in the system. In the case of the use of Bluetooth technology, the smartphone will be connected to the HC-05 module, which will be used for the transmission of data between the mobile application and the Arduino microcontroller. In the case of the use of the Wi-Fi communication protocol, the smartphone will be connected to the hotspot created by the ESP8266 module, which will provide a web interface for the creation of the emergency message. The message will contain crucial information, which will include an SOS message, the identity of the sender, as well as the nature of the emergency being experienced. The creation of the message has been simplified, enabling an illiterate person to create an emergency message at a critical moment. The message will be sent using the mobile device, which will be transmitted to the communication module.

3. Data Processing and Transmission

The Arduino Uno receives the message sent from the Bluetooth or Wi-Fi module and then takes the responsibility to process the data. This is an important step to ensure that the data sent is correct and in the right format to enable long-distance communication. The Arduino processes the data and formats it in a way that can be easily understood at the receiving end. This involves formatting the data string and coding it in a way that reduces the chances of error during transmission. Once this process is done, the data is sent to the Zigbee (HC-12) module via the Arduino module. The Zigbee module uses wireless communication to enable the transmission of the message to a long distance via radio frequency signals. This form of communication is independent of mobile network and internet connectivity. This form of communication can be relied upon during disaster times when communication is not possible. The use of Zigbee helps to save energy while transmitting reliable signals in harsh conditions.

4. Message Reception and Decoding

The Zigbee (HC-12) module, at the receiver unit, is always on the lookout for any signal. This helps it to detect any data that is being transmitted. Once it detects a signal, it captures the data and sends it to the Arduino Uno. The Arduino Uno decodes the data that it receives. This involves decoding the data by understanding who sent the message, the type of emergency, and any other information that the sender of the message wants to convey. This decoding of the message plays a crucial role in ensuring that the information that one receives is accurate. By decoding the message, it becomes easy to communicate the information to the rescue team

without any confusion. This plays a crucial role in ensuring that the effectiveness of the communication system is achieved.

5. Alert and Output Display

Once the message has been successfully decoded, the system proceeds to notify the rescue personnel. This is done in both visual and audible forms. The Arduino sends the decoded message to the 16x2 LCD display. This message is presented in a clear and readable format. This allows the rescue team to understand the nature of the emergency and respond accordingly. At the same time, a buzzer is also connected to the Arduino. This produces a sound that draws immediate attention to the message. This is particularly important in noisy environments. This stage in the system enhances the overall efficiency of the system. This is because no message is left unattended to.

6. Overall System Operation

The overall operation of this system ensures that seamless communication takes place between the user and the rescue team. All steps of this workflow are interconnected, ensuring that communication takes place smoothly without any interruption. The combination of using Bluetooth and Wi-Fi for inputting messages, along with Zigbee for transmitting messages over long distances, ensures that this communication system is efficient. This system has been designed to be used for a longer time with low power consumption. Moreover, this system has been designed to be used for two-way communication, as the receiver unit will be able to send messages to the transmitter unit using this system. This will allow better decision-making during such operations. Overall, this workflow has been designed to make this communication system efficient, providing a reliable means of communication during such disaster situations when communication networks are not available.

CHAPTER 5
HARDWARE COMPONENTS

5.1 Arduino Uno

In this project, the Arduino Uno microcontroller is the main controller that oversees the functioning of all other hardware components. The microcontroller is responsible for processing the information, communicating with the modules, and ensuring that there is the transfer of information to other devices. The microcontroller is programmed using the Arduino IDE to receive and send emergency information, status information, and communication information even without internet connectivity. The microcontroller is reliable in processing the information, making it suitable for prototyping in emergency situations.



Fig 5.1 Arduino Uno

The microcontroller's digital and analog pins are used to connect to other hardware components such as the LCD display, ESP8266, and Zigbee module. During operation, the microcontroller is responsible for ensuring timely transmission of messages to the victims and rescue teams using the wireless communication module. The microcontroller is easy to use,

making it suitable for operation in emergency situations where there may be a lack of power sources.

In conclusion, the Arduino Uno microcontroller is the main controller that oversees the functioning of the emergency communication system. The microcontroller is efficient in processing information and ensuring timely transmission of information to the victims and rescue teams

5.2 ESP8266

The ESP8266 Wi-Fi module is utilized in this project to facilitate the process of wireless data communication between Arduino devices or with mobile applications. This module helps to facilitate the establishment of a local wireless network that does not require internet access to send emergency messages or information related to the victims to the rescue team. The ESP8266 module is utilized in the project to facilitate the establishment of a hotspot that enables the establishment of a connection between smartphones and the module to send messages.

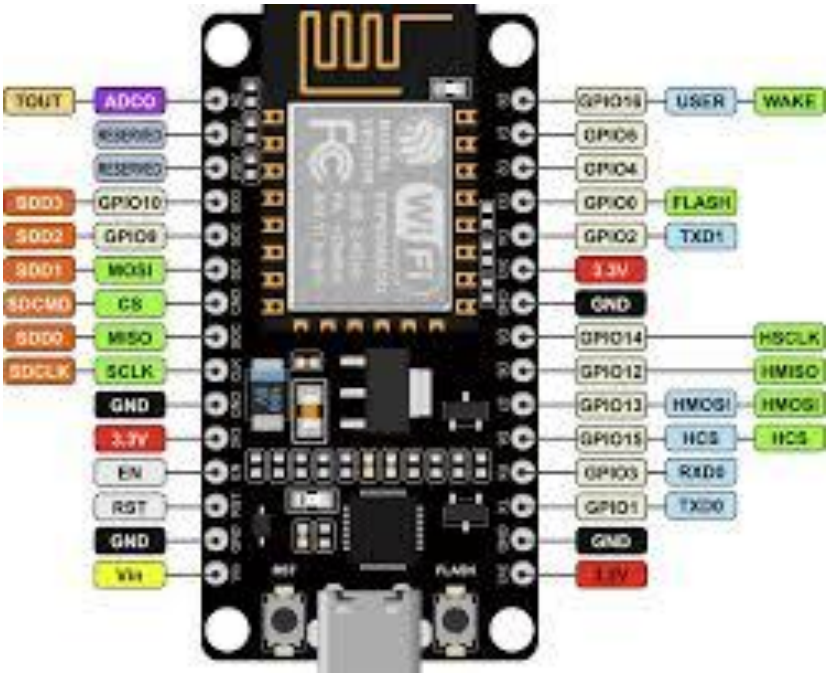


Fig 5.2 ESP8266

The module is utilized to facilitate the sending of data packets in a secure manner using the Arduino IDE. The module is compact in size and uses minimal power, making it easy to utilize in the real world. The module helps to facilitate the establishment of an ad-hoc network to enable connectivity between various devices in the area.

Table 5.1 ESP8266

ESP8266 Wi-Fi Module	Arduino Uno	Description
VCC	3.3V	Power supply (use 3.3V only)
GND	GND	Common ground
TX	RX(pin 0)	Data sent from ESP8266 to Arduino
RX	TX(pin 1)	Data sent from Arduino to ESP8266 (use voltage divider for 3.3V logic)
EN	3.3V	Chip enable pin to activate ESP8266

5.3 LCD Display

The LCD Display (16x2) is used to display system information to users or rescuers. It displays messages such as device status, network connections, alert messages, and power levels. This display is the primary output device for users in the field. Users can easily read communication information.

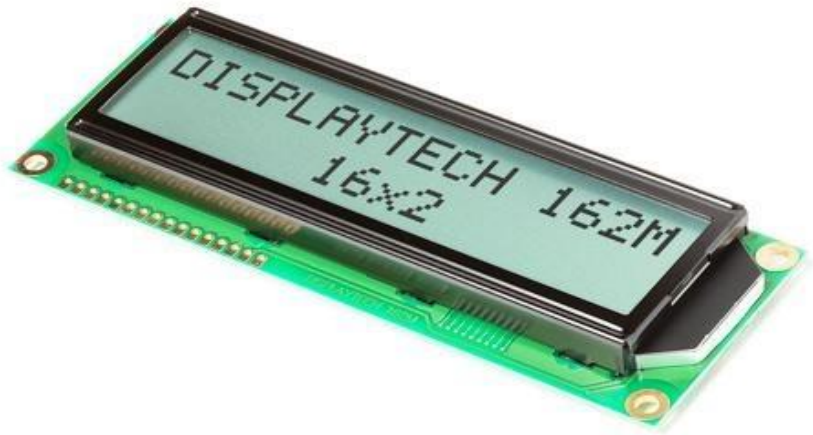


Fig 5.3 LCD Display

The display is connected to the Arduino board using digital pins. Libraries are also used to send text messages to the display. During the rescue operation, it displays messages such as “Message Sent,” “Network Connected,” and “Low Battery.”

The display is beneficial to users. It displays information in an easily readable format. The display also has a backlight that allows users to read messages in dark environments. This display is also portable and light in weight.

Table 5.2 LCD

LCD Display	Arduino Uno	Description
VSS	GND	Ground connection
VDD	5V	Power supply for LCD
V0	Potentiometer Output	Adjust contrast
RS	Digital Pin 7	Register select control
RW	GND	Set to write mode
EN	Digital Pin 8	Enable signal for data latch
D4	Digital Pin 9	Data bit 4
D5	Digital Pin 10	Data bit 5
D6	Digital Pin 11	Data bit 6
D7	Digital Pin 12	Data bit 7
A (LED+)	5V (via 220Ω resistor)	Backlight power
K (LED–)	GND	Backlight ground

5.4 HC-12 Zigbee Module

The Zigbee module is one of the main components that is utilized for long-distance wireless communication between multiple devices. It uses serial communication to send emergency messages between the rescue team and the victims without the use of the internet or GSM technology.

Table 5.3 Zigbee Module

HC-12 Zigbee Module	Arduino Uno	Description
VCC	5V	Power supply
GND	GND	Common ground
TX	Digital pin 10	Data Transmission
RX	Digital Pin 11	Data reception

The Zigbee module is capable of transmitting data up to 1 kilometer in open areas. This is beneficial for disaster areas where communication is down. In this project, the Zigbee module is connected to the Arduino Uno using UART communication to send and receive data.
The Zigbee module is reliable for use in emergency situations due to its low power consumption and reliable communication. It ensures that the messages are sent correctly even in harsh environments.

5.5 Li-ion Battery

The entire system is powered by a Li-ion Battery. This allows the system to be portable and to operate in situations when there is a power outage. This is particularly common in disaster situations. Since the system is to operate in the field, where there might be no power supply, the battery serves as the primary power source to power the Arduino and other modules. This ensures that all the modules operate effectively. In the project, the Li-ion Battery is a rechargeable one. It is connected to the Arduino system through a voltage regulator circuit. This ensures that the battery is not subjected to any form of overvoltage. The battery is also very light in the sense that it has high density. This allows the system to be portable.



Fig 5.4 Li-ion Battery

These safety features will ensure that this type of power system is not only reliable but also durable to ensure that it is used for a longer period without the need for frequent maintenance or replacement. The Lithium-Ion (Li-ion) battery is an important component of this project. The battery will be used as a source of power to ensure that the system is operational at all times in any environment or situation, including emergency or disaster situations. The environment in which the system is expected to operate is one where access to conventional

sources of power is not available. In an environment where access to conventional sources of power is not available, the Li-ion battery will provide the source of power to operate the Arduino Uno system appropriately. The function of the Li-ion battery is to ensure that the system is operated appropriately to send important information even in situations where there is no power or electric power is not available.

The Li-ion battery has the ability to power portable electronic devices because of its high energy density, low self-discharge, and rechargeable properties. The Li-ion battery has properties that make it useful for powering electronic devices that need a battery with a long lifespan, light weight, and efficiency. In this project, the Li-ion battery will serve as a power source, which will provide a constant voltage supply to the Arduino Uno, as well as other components like the HC-12 transmitter, ESP8266 Wi-Fi, and other sensors. The Li-ion battery has a high energy density, which allows it to store a lot of power while being light, thus portable during rescue missions.

Table 5.4 Li-ion Battery

Li-ion Battery	Arduino Uno	Description
+ve Terminal	VIN	Power supply input to arduino
-ve Terminal	GND	Common ground

5.6 Voltage Regulator

The Voltage Regulator is one of the most crucial components in the project. This is because it ensures that all the devices connected to it operate at safe voltage levels. The emergency communication system requires a constant voltage supply to ensure proper functioning. The voltage regulator is essential in ensuring that the voltage level output by the Li-ion battery is constant at 5V or 3.3V. This ensures that the components in the system are protected against any electrical damages.

In the project, the voltage regulator is very crucial in ensuring that all the components in the system operate at the correct voltage levels. The Arduino Uno board operates at 5V voltage levels. However, the ESP8266 module operates at 3.3V voltage levels. Connecting these components to unregulated voltage levels may cause overheating and malfunction. The voltage regulator ensures that the voltage output levels match the voltage levels required by the

components. This ensures that all the components operate at stable voltage levels even when the voltage output levels vary.

The voltage regulator is positioned between the battery and the system to ensure performance. The voltage regulator ensures that there is no sudden drop in voltage that would cause the system to fail. The constant performance by the voltage regulator ensures that the system is durable and efficient. This makes the system reliable in case of disasters when communication is crucial.

Table 5.5 Voltage Regulator

Voltage Regulator	Arduino Uno	Description
Input	Battery +ve	Input power
Output	5v pin	Regulated output for Arduino and other modules
GND	GND	Common ground connection

5.7 Supporting Components

Other than these hardware components, there are some components which are used to operate the system properly.

Breadboard

The breadboard is used to test the circuit without actually soldering the components.

Connecting Wires (Jumper Wires)

Jumper wires are used to make connections with different components, like Arduino, etc.

Resistors

Resistors are used to restrict the flow of current and divide the voltage. For example, a resistor with a value of 220 ohms is used to enable the backlight of the LCD display, while voltage division resistors are used to interface with ESP8266.

Capacitors

Capacitors are used to filter the voltage supply, which enables the smooth flow of voltage.

Potentiometer

Potentiometer is used to adjust the contrast of the LCD display so that the text displayed on the screen is visible.

Switch

Switch is used to control the voltage supply to the circuit so that it is easy to switch ON/OFF.

5.8 Conclusion

The Short Range Offline Communication System offers a reliable solution in circumstances when conventional means of communication do not work. The integration of components like Arduino Uno, ESP8266, Zigbee module, and Li-ion battery in the system offers a means of communication between the victims and rescue teams even in circumstances when there is no Internet or GSM. The system is portable, energy-efficient, and can be used in disaster management situations. The system proves the potential of embedded systems in ensuring reliable communication in critical circumstances.

In conclusion, the Short Range offline Communication System is a giant leap towards enhancing the reliability of communication in critical circumstances. The system offers a reliable, portable, energy-efficient, and cost-effective means of communication. The system proves the potential of embedded technology in ensuring reliable communication in critical circumstances. The system also proves the potential of wireless communication in ensuring reliable communication in critical circumstances.

CHAPTER 6
IMPLEMENTATION

The implementation of the Short-Range Offline Communication System focuses on developing a secure and reliable communication link between various devices without relying on any of the existing wireless technologies such as Wi-Fi or cellular networks. The system is designed by utilizing an embedded microcontroller-based platform that communicates with short-range wireless technologies such as Bluetooth, RF, or ZigBee. The microcontroller acts as the primary processing device of the system, responsible for controlling the data transmission and reception between various devices. The short-range wireless modules are designed to operate within the license-free band of the ISM frequency range.

This helps to provide wireless communication at a lower cost. The integration of these modules helps the system to operate independently without relying on any of the existing wireless networks. The user interface of the system is designed to be simple and user-friendly. This interface may be provided by developing a mobile application. The user sends a message to the system, which is converted into digital data by the microcontroller. This data is transmitted to the microcontroller by utilizing Bluetooth or WiFi modules. The microcontroller processes this data by applying encoding techniques to minimize errors during data transmission.

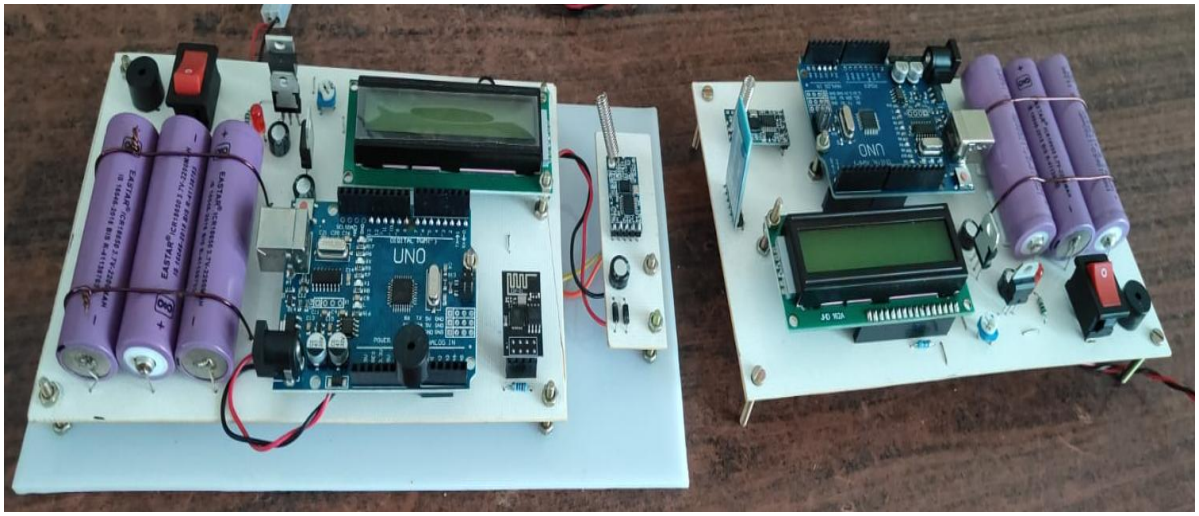


Fig. 6.1 Complete Circuit

On the receiver side, the signal transmitted is picked up by the communication module, which forwards it to the microcontroller. The signal is decoded and displayed on a device, like an LCD

screen or a mobile phone. The design is portable and consumes less power, which is achieved by running it on batteries. In addition, the design has incorporated some important features, like sending acknowledgement signals, handling collisions, and validating the data, to make the communication process even better. Moreover, the hardware is housed in a casing, which is compact and rugged, making it easier to carry and operate in any environment. The design has focused on saving power, reducing hardware complexity, and ensuring privacy. During the implementation, the design has been tested in various environmental conditions, like indoor, outdoor, and even in an environment where there is an obstruction. The parameters on which the design has been tested include the range, delay, and signal stability. Based on the implementation, it is safe to conclude that the design has been successful in achieving an offline communication system, which is quite efficient and can operate independently without any support from any existing network infrastructure.

6.1 RESULTS

The experimental results of the Short-Range Offline Communication System have proven that the suggested system indeed works successfully as an efficient standalone communication system. During the experiment, it was successfully proven that the two units of the communication system were capable of communicating with each other by sending and receiving text messages without relying on the internet or cellular network.

The suggested system successfully proved that it could be used for real-time communication, where the messages sent through the mobile application were successfully transmitted to the other device within an insignificant period of time.

Step-wise Result Observation:

Message Transmission from Victim to Rescuer

The first part of the result observation shows the process of communication that is initiated by the victim. The whole process of the suggested system's working is clearly explained step by step as follows:

Step 1: Message Input by Victim

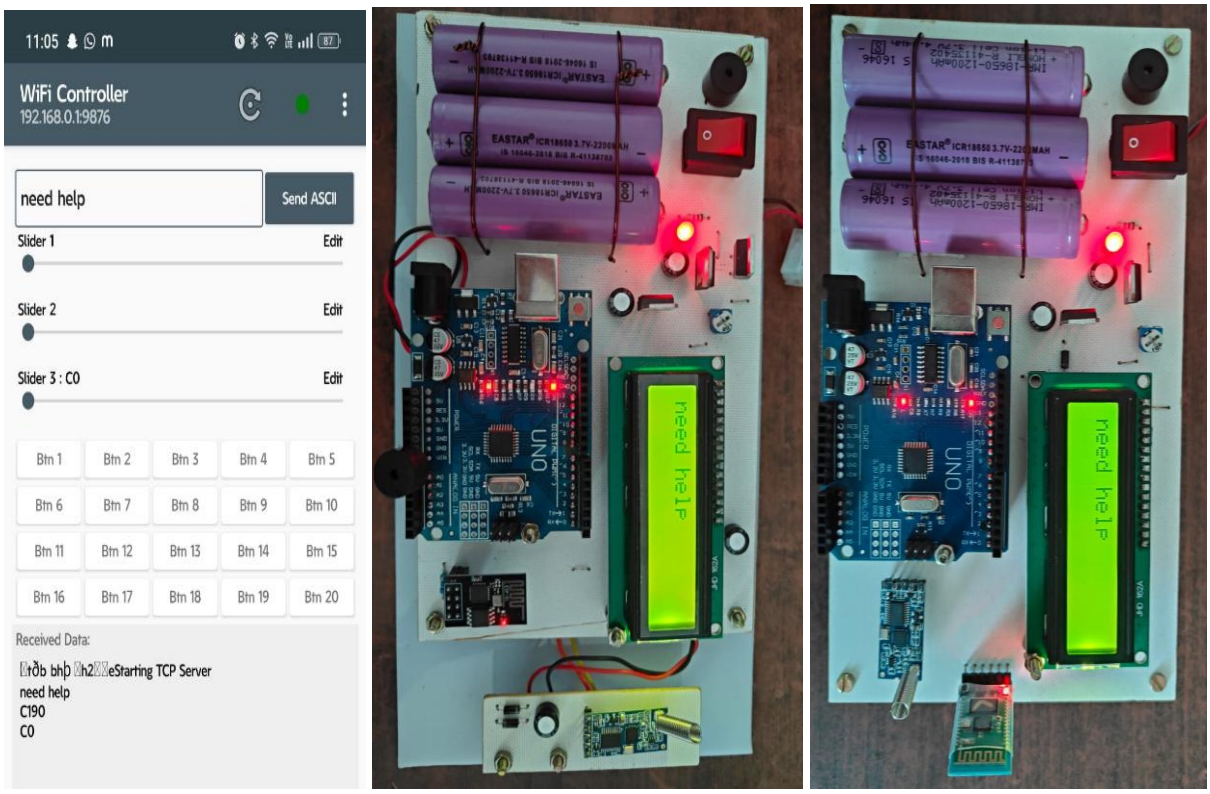
- The victim sends a message such as "need help" by using a mobile application known as the Wi-Fi controller application.
- The mobile device sends the message to the embedded system via the Wi-Fi module.

Step 2: Transmission to Victim Node

- The typed message is transmitted wirelessly from the mobile application to the Wi-Fi module connected to the Arduino board.

Step 3: Processing at Victim Node

- The Arduino processes and decodes the message.
- The message is immediately displayed on the LCD screen at the victim node.



.**Fig.6.2**. When Victim sends the message

Step 4: Transmission via Zigbee Module

- The Arduino at the victim node sends the same message to the Zigbee module.
- The Zigbee module transmits the message wirelessly using RF communication to the rescuer node.

Step 5: Reception at Rescuer Node

- The Zigbee module at the rescuer side receives the transmitted message.
- The Arduino at the rescuer node decodes the received data.

Step 6: Display and Alert at Rescuer Side

- The message is displayed on the LCD screen at the rescuer node.

- Simultaneously, a buzzer is activated to alert the rescuer that a message has been received from the victim.

Result Observation

The display of the message on both nodes along with the activation of the buzzer confirms reliable one-way communication.

Response from Rescuer to Victim

The second stage in the flow diagram shows reverse communication between the victim and rescuer. The steps involved in this stage are given below:

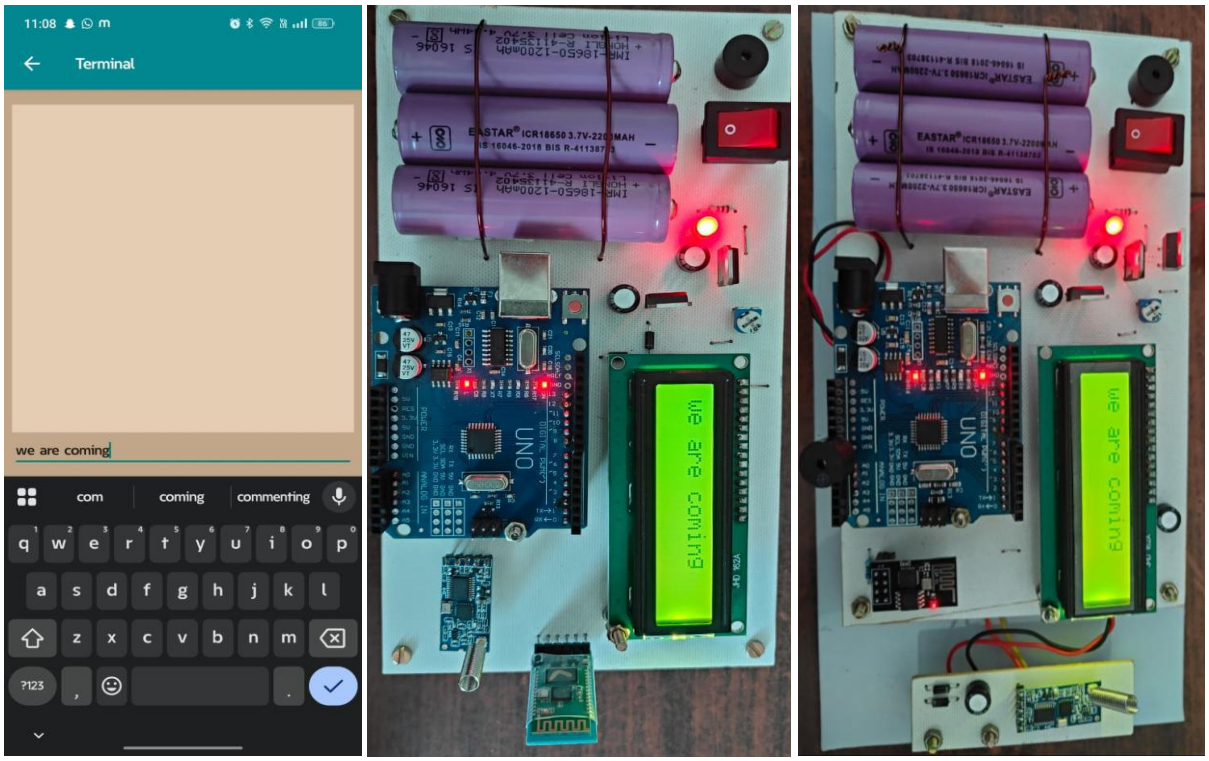


Fig.6.3. Rescuer responds to the message

Step 1: Message Input by Rescuer

- The rescuer uses a mobile application connected via Bluetooth to type a response message such as “we are coming.”
- The mobile device sends a request to establish a connection with the Bluetooth module connected to the rescuer node, which is the HC-05 device.

Step 2: Transmission to Rescuer Node

- The mobile application sends the message to the Bluetooth device.
- The Arduino receives the message at the rescuer node.

Step 3: Processing at Rescuer Node

- The Arduino processes the response message.
- The response message is displayed on the LCD of the rescuer node.

Step 4: Transmission via Zigbee Module

- The Arduino sends the response message to the Zigbee device.
- The Zigbee device sends the response message wirelessly to the victim node.

Step 5: Reception at Victim Node

- The Zigbee module at the victim's end receives the transmitted response.
- The Arduino receives the decoded message.

Step 6: Display and Alert at Victim Side

- The received response message is displayed on the LCD screen at the victim's end.
- The buzzer is activated to alert the victim that a response has been received.

Result Observation

The successful display of messages on both nodes confirms that reliable communication has taken place from the victim to the rescuer.

6.2 CONCLUSION

The design and implementation of the "Short Range Offline Communication System" successfully prove that reliable communication is possible without internet or cellular connectivity. This system successfully enables two-way communication between victims and rescuers using Wi-Fi, Bluetooth, and Zigbee communication modules.

The results obtained from this experimentation show that messages sent from the victim's end are successfully received, displayed, and alerted at the rescuer's end, and vice versa. The use of LCD displays ensures effective visualization of messages, while the buzzer ensures immediate alerting.

This project is extremely beneficial in situations where traditional means of communication are impossible, like in disaster scenarios, rescue operations, underground, and remote locations. The project offers a safe and self-sustaining means of communication, ensuring that information is shared irrespective of the current infrastructure. As far as technical knowledge is concerned, this project has offered valuable insights into the world of embedded systems, wireless communications, and optimal means of data transfer.

6.3 FUTURE SCOPE

Although the current system is functioning properly in terms of short-range offline communication, there are many opportunities to enhance the future scope to increase the efficiency, scalability, and security of the current system.

- **Integration of Long-Range Communication Technologies**

  One of the most important improvements is the integration of long-range communication technology, which may include LoRa technology. LoRa is a type of module that is used in long-range communication, where the range may be up to several kilometers. The most important advantage is that this module would increase the area coverage while maintaining low power consumption.

- **Implementation of Mesh Networking**

  Mesh networking is another important improvement to the current system, which would transform the current system into a decentralized communication network. In this type of network, there is a possibility to use several devices, which would operate like nodes to communicate with each other. In this case, even if the device is not functioning properly, the message would be sent to the intended device.

- **Enhanced Security and Data Protection**

  Future versions of the system can incorporate advanced encryption techniques such as AES or RSA to ensure secure data transmission. This would protect sensitive information from unauthorized access or interception. Additionally, implementing authentication mechanisms between devices can ensure that communication occurs only between trusted users, thereby improving overall system security and reliability.

- **Implementation of Mesh Networking**

  Introducing mesh networking can transform the system into a decentralized communication network. In this approach, multiple devices can act as nodes that relay messages to one another, allowing communication beyond direct range. Even if one node fails, the message can still reach its destination through alternate paths, thereby improving system reliability and fault tolerance. This is particularly useful in emergency situations where infrastructure.

- **Support for Multimedia and Emergency Alerts**

  The system may be extended to accommodate other multimedia messages such as short audio alerts, alert tones, and emergency alerts. This feature is helpful in emergency situations where typing may not be feasible. Quick alert buttons may be provided to enable the user to send emergency alerts instantly. Such alerts may be helpful in emergency situations to enable timely response to emergency situations.

CHAPTER 7

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